

Hypopigmentation, Organomegaly, and Delayed Myelination and Development (HOD)

Comprehensive Disease Knowledge-Base Report

Answer in brief: HOD is an ultra-rare, dominantly-acting Mendelian multisystem disorder caused by *de novo* **gain-of-function** missense variants in **CLCN7**, which encodes the lysosomal $2\text{Cl}^-/1\text{H}^+$ antiporter **CLC-7** (obligate partner: OSTM1). Excessive CLC-7 activity **hyper-acidifies lysosomes and lysosome-related organelles (melanosomes)**, causing generalized hypopigmentation/oculocutaneous albinism, lysosomal storage with hepatosplenomegaly (organomegaly), and CNS involvement (delayed myelination, developmental delay). It is mechanistically the *mirror image* of CLCN7 **loss-of-function/dominant-negative** disease (osteopetrosis), and reported HOD patients characteristically **lack osteopetrosis**.

1. Disease Information

- **Overview:** HOD ("Hypopigmentation, Organomegaly, and Delayed Myelination and Development") is a multisystem congenital disorder combining diffuse skin/hair/eye hypopigmentation (albinism-like), organomegaly (hepatosplenomegaly), and neurodevelopmental impairment with delayed CNS myelination. It is a **lysosomal storage phenotype** driven by abnormal lysosomal acidification.
- **Key identifiers:**
- **OMIM #618541** — "Hypopigmentation, Organomegaly, and Delayed Myelination and Development."
- **Gene:** CLCN7 (OMIM *602727), HGNC:2023, locus 16p13.3.
- **Mondo:** MONDO:0032781 (HOD; per Mondo cross-reference to OMIM 618541 — to be confirmed against current Mondo release).

- **Orphanet / ICD-11:** No dedicated Orphanet code identified; closest ICD-11 grouping is under lysosomal/metabolic and pigmentary disorders. ICD-10: no specific code (would map broadly to E75.- lysosomal / E70.3 albinism categories). *Not available as a discrete code.*
- **MeSH:** No dedicated MeSH descriptor; relevant terms: "Lysosomal Storage Diseases", "Albinism", "CLCN7 protein, human".
- **Synonyms / alternative names:** "CLCN7-related gain-of-function lysosomal disorder"; "CLCN7 gain-of-function hypopigmentation and lysosomal storage disease"; informally "HOD syndrome." Related descriptor in literature: "multisystem disorder associated with a pathogenic variant in CLCN7 in the absence of osteopetrosis."
- **Data source:** Information is **individual-patient / small-case-series derived** (fewer than ~10 reported probands), supported by disease-level functional/model-organism resources. This is not an aggregated EHR-scale entity.

2. Etiology

- **Primary cause (genetic):** Heterozygous **de novo gain-of-function (GoF)** missense variants in **CLCN7**. Index variant **p.Tyr715Cys (Y715C)** in the C-terminal cystathionine- β -synthase (CBS) domain; the original AJHG report (Nicoli et al., 2019;

P 31155284

described additional de novo variants (e.g., p.Tyr99Cys/Y99C in the transmembrane domain) with the same functional signature. Functional testing shows **increased outward currents** → lysosomal hyperacidity

(

P 39056574

).

P 31155284

- **Genetic risk factors:** The causal variants are the sole known determinant; because they are *de novo* and dominant-acting, there is no meaningful susceptibility-locus/modifier background documented. Modifier genes and compensatory pathways are hypothesized but unproven

(

P 39056574

- **Environmental risk factors:** None identified. This is a monogenic disorder; onset is congenital and independent of exposures, lifestyle, or sex.

- **Protective factors:** None described. (Hypothetically, lowering ClC-7 activity/lysosomal acidity could be protective — see Treatment.)
 - **Gene–environment interactions:** Not applicable / none reported.
-

3. Phenotypes

All frequencies are qualitative given the very small case count (individual-patient evidence).

Phenotype	Type	HPO term	Onset	Severity	Frequency
Generalized hypopigmentation of skin/hair	Physical/pigmentary	HP:0001010 (Hypopigmentation of the skin), HP:0002286 (Fair hair)	Congenital	Moderate–severe	Core/near-constant
Oculocutaneous/ocular albinism features (iris hypopigmentation, nystagmus, reduced visual acuity)	Clinical sign	HP:0007730 (Ocular albinism), HP:0000639 (Nystagmus)	Congenital/neonatal	Variable	Common
Hepatomegaly / splenomegaly (organomegaly)	Clinical sign	HP:0002240 (Hepatomegaly), HP:0001744 (Splenomegaly), HP:0003271 (Visceromegaly)	Infancy	Moderate	Core
Delayed CNS myelination (white-matter)	Imaging/lab	HP:0012448 (Delayed CNS myelination)	Infancy	Variable	Core
Global developmental delay / intellectual disability	Behavioral/cognitive	HP:0001263 (Global developmental delay), HP:0001249 (Intellectual disability)	Infancy	Mild–severe, variable	Core
Hypotonia / motor delay	Neurological sign	HP:0001252 (Hypotonia)	Infancy	Variable	Common
Hypogammaglobulinemia / immune involvement	Lab abnormality	HP:0002090? →HP:0004313 (Decreased circulating antibody level)	Infancy	Variable	Reported (Lee 2024)
Vacuolated lymphocytes / lysosomal storage cells	Lab/histology		Congenital	—	Reported

Phenotype	Type	HPO term	Onset	Severity	Frequency
		HP:0001922 (Abnormal blood-cell morphology)			
Coarse/dysmorphic features (variable)	Physical	HP:0000280	Congenital	Mild	Variable

- **Progression:** Neurodevelopmental features are static-to-slowly-progressive; underlying storage biology in models is progressive/neurodegenerative.
- **Quality-of-life impact:** Substantial — visual impairment (albinism, nystagmus), developmental disability, and immune susceptibility affect daily function. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare entity.
- **Notably ABSENT:** osteopetrosis/increased bone density (distinguishes HOD from CLCN7 LoF disease).

4. Genetic / Molecular Information

- **Causal gene:** CLCN7 (HGNC:2023; NCBI Gene 1186; UniProt **P51798**, CLCN7_HUMAN), 16p13.3. Encodes CLC-7, a voltage-gated CLC-family Cl⁻/H⁺ **antiporter** (2Cl⁻:1H⁺ stoichiometry).
- **Pathogenic variants (HOD, GoF):**
- **c.296A>G, p.Tyr99Cys (Y99C)** — transmembrane domain (original de novo cases, Nicoli 2019).
- **c.2144A>G, p.Tyr715Cys (Y715C)** — C-terminal CBS domain (Nicoli 2019; Lee 2024, [P 39056574](#)).
- **Variant type/class:** missense (single amino-acid substitutions); **germline de novo**, not somatic.
- **Classification (ACMG/AMP):** pathogenic/likely pathogenic (de novo, functionally validated, absent from population databases).
- **Allele frequency:** Absent from **gnomAD** (private de novo). CLCN7 is highly constrained.
- **Functional consequence: Gain of function** — increased Cl⁻/H⁺ transport → lysosomal hyperacidification and accelerated/enhanced currents

(
P 39056574
P 31155284
P 38136669
).

- **Contrast — CLCN7 osteopetrosis variants:** Dominant-negative (ADO2/Albers-Schönberg, e.g., p.Gly213Arg) or biallelic loss-of-function/NMD (ARO, e.g., p.Met59Asnfs*8) — opposite functional direction, different disease

(
P 11741829
P 39994654
P 17936098
).

- **Modifier genes:** OSTM1 (obligate β -subunit) is essential for CLC-7 function; unproven as a HOD modifier. Others hypothesized

(
P 39056574
).

- **Epigenetic / chromosomal abnormalities:** None reported; disease is single-nucleotide/missense, not structural.

Ontology suggestions: Gene HGNC:2023 (CLCN7); protein UniProt P51798; partner OSTM1 (HGNC:16800, UniProt Q86WC4).

5. Environmental Information

- **Environmental factors:** None. Congenital monogenic disorder.
- **Lifestyle factors:** Not applicable.
- **Infectious agents:** None causal. (Note: hypogammaglobulinemia may predispose to recurrent infections as a *consequence*, not a cause.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **de novo gain-of-function missense variant** in *CLCN7* (e.g., Y99C, Y715C) alters ClC-7 gating/coupling → **leads to** increased $2\text{Cl}^-/1\text{H}^+$ antiporter activity (demonstrated by increased outward currents in *Xenopus* oocytes;

P 39056574

P 31155284

2. Enhanced Cl^- counter-ion flux **provides greater charge shunt** for the V-ATPase → **results in over-acidification (hyperacidity) of lysosomes** and lysosome-related organelles (demonstrated: "lysosomal hyperacidity";

P 39056574

3. Abnormally low luminal pH **disrupts pH-optimal lysosomal hydrolases and membrane trafficking** → **impairs autophagic/degradative clearance** (demonstrated in cell models of GoF variants;

P 38136669

→ **results in accumulation of undigested storage material and cytoplasmic vacuolization** (demonstrated;

P 39056574

4. **Branch A — Melanosomes (pigment):** melanosomes are lysosome-related organelles whose melanin synthesis (tyrosinase) is exquisitely pH-sensitive. Hyperacidification **impairs melanin production** → **generalized hypopigmentation / albinism-like phenotype** (inferred mechanism, supported by melanosome-pH biology and the albinism phenotype in Nicoli 2019 / Polovitskaya 2024).
5. **Branch B — Reticuloendothelial storage:** storage-material accumulation in hepatic/splenic macrophages and parenchyma → **hepatosplenomegaly / organomegaly** (inferred from lysosomal-storage biology; consistent with organomegaly in reported patients).
6. **Branch C — CNS:** neuronal and glial (oligodendrocyte, microglial) storage and dysfunction → **impaired myelination and neuronal function** → **delayed myelination + developmental delay** (supported by *Clcn7*-KO mouse neurodegeneration/NCL-like storage,

P 15706348

and microglial phagocytic failure,

P 38294065

7. **Branch D — Immune:** impaired lysosomal/endosomal processing in B-lineage/antigen-processing cells → **hypogammaglobulinemia** (reported;

P 39056574

8. Chronic storage/vacuolization in neurons and glia **is inferred to drive progressive neurodegeneration** (from model organisms; NCL-like subunit-c and lysosomal-enzyme accumulation,

P 15706348

Category detail

- **Molecular pathways:** Lysosomal ion homeostasis / V-ATPase-coupled acidification; autophagy–lysosome pathway; mTORC1 lysosomal signaling (inferred). Reactome "Ion channel transport"; KEGG "Lysosome" (map04142).
- **Cellular processes:** Autophagic clearance (impaired), lysosomal degradation, phagocytosis (microglial clearance of apoptotic cells/ β -amyloid impaired;

P 38294065

melanogenesis.

- **Protein dysfunction:** CLC-7 GoF (altered gating/uncoupling of Cl^-/H^+ transport); obligate complex with OSTM1 — OSTM1 is required for CLC-7 stability (CLC-7 protein falls >90% in Ostm1-null grey-lethal mice;

P 16525474

and the OSTM1 dimer shields luminal CLC-7 (cryo-EM;

P 32749217

- **Disease-family paradigm:** HOD belongs to a broader class of **endolysosomal CLC gain-of-function neurodevelopmental disorders** — a recurrent GoF mutation in the paralog **CLCN6 (CLC-6)** causes a neurodevelopmental/neurodegenerative disorder via the same endolysosomal-ion-homeostasis logic (Polovitskaya et al. 2020,

P 33217309

and endolysosomal CLC dysfunction generally produces NCL-like storage and neurodegeneration (review

P 33708769

CLC-3 deficiency NCL model

P 12059962

- **Metabolic changes:** Accumulation of NCL-type storage (subunit c of mitochondrial ATP synthase) and lysosomal enzymes

P 15706348

- **Immune involvement:** Hypogammaglobulinemia; microglial dysfunction.
- **Tissue-damage mechanism:** Storage-driven cellular dysfunction and neurodegeneration (astrogliosis, microglial activation in models).
- **Biochemical abnormality:** Dysregulated lysosomal pH set-point (hyper-acidic).

Melanosome-pH sub-mechanism (hypopigmentation branch, supporting evidence): Melanosomal luminal pH is a master regulator of melanin synthesis; **tyrosinase (the rate-limiting melanogenic enzyme) is minimally active in an acidic environment** (Miao et al. 2019,

P 31214276

and melanosomal pH is normally set by V-ATPase proton pumping balanced by ion transporters (SLC45A2/OCA4, OCA2 P-protein, SLC24A4/5) (Cheli et al. 2009,

P 19389708

CLC-7 supplies the Cl⁻ counterion that enables V-ATPase-driven organelle acidification (Mindell 2012,

P 22335796

Thus **CLC-7 GoF → melanosome over-acidification → tyrosinase suppression → hypopigmentation**. This places HOD within a broader class of "ion-transport / organelle-pH" albinism syndromes — e.g., gain-of-function TPC2 p.R210C albinism

(
P 36641477
 and V-ATPase-deficiency oculocutaneous albinism in zebrafish
 (
P 18836173
).

GO/CL suggestions: GO:0007042 (lysosomal lumen acidification), GO:0055085 (transmembrane transport), GO:1902600 (proton transmembrane transport), GO:0006914 (autophagy), GO:0006622 (protein targeting to lysosome), GO:0006583 (melanin biosynthetic process from tyrosine), GO:0042470 (melanosome). Cellular component: GO:0005765 (lysosomal membrane), GO:0042470 (melanosome). Cell types: CL:0000540 (neuron), CL:0000128 (oligodendrocyte), CL:0000129 (microglial cell), CL:0000148 (melanocyte), CL:0000091 (Kupffer cell), CL:0000236 (B cell).

CHEBI suggestions: CHEBI:17594 (melanin), CHEBI:17996 (chloride), CHEBI:24636 (proton/H⁺), CHEBI:15377 (water), CHEBI:18059 (lipid, storage material context). Key protein/enzyme: tyrosinase (EC 1.14.18.1, UniProt P14679).

7. Anatomical Structures Affected

- **Organ level:** Skin & hair (pigment), eyes (iris/retina), liver & spleen (organomegaly), central nervous system (white matter/brain), immune system (B-cell compartment). Bone is **spared** (no osteopetrosis).
- **Body systems:** Integumentary, nervous, hepatobiliary/reticuloendothelial, visual/ophthalmic, immune.
- **Tissue/cell level:** Melanocytes (CL:0000148); neurons (CL:0000540) and oligodendrocytes (CL:0000128, myelination); microglia (CL:0000129); Kupffer cells/macrophages (CL:0000235/CL:0000091); B lymphocytes (CL:0000236).
- **Subcellular:** Lysosome (GO:0005764) and lysosomal membrane (GO:0005765); lysosome-related organelles / melanosomes (GO:0042470); late endosome (GO:0005770); autophagosome (GO:0005776).
- **Localization / lateralization:** Systemic and **bilateral/symmetric** (pigmentary, hepatosplenic, CNS white-matter changes are generalized).

UBERON suggestions: UBERON:0002097 (skin of body), UBERON:0002107 (liver), UBERON:0002106 (spleen), UBERON:0000955 (brain), UBERON:0002316 (white matter), UBERON:0000970 (eye).

8. Temporal Development

- **Onset:** Congenital / neonatal (hypopigmentation evident at birth; developmental delay and organomegaly in infancy).
 - **Onset pattern:** Chronic, insidious.
 - **Progression:** Neurodevelopmental delay is largely static-to-slowly-progressive in reported patients; model-organism data indicate an underlying **progressive neurodegenerative** storage process.
 - **Course:** Chronic, lifelong. No remission described.
 - **Critical periods:** Early infancy (myelination window) is the key period of CNS vulnerability and the plausible window for intervention.
-

9. Inheritance and Population

- **Epidemiology:** Ultra-rare; **<10 reported cases worldwide** (individual case reports/small series). Prevalence/incidence not estimable (effectively unknown; well below 1/1,000,000).
- **Inheritance:** **Autosomal dominant, de novo** (sporadic). Not inherited from parents in reported cases.
- **Penetrance:** Presumed complete for the specific GoF alleles (all reported carriers affected).
- **Expressivity:** Variable (developmental severity, immune involvement differ between patients).
- **Anticipation / mosaicism / founder effects / consanguinity / carrier frequency:** Not applicable (de novo dominant; no carrier state, no founder or consanguinity role). Germline mosaicism is theoretically possible for recurrence risk counseling but unreported.
- **Population demographics:** No ethnic predilection; cases reported across populations (e.g., a Taiwanese proband,

P 39056574

No sex bias expected (autosomal). Age distribution: pediatric.

10. Diagnostics

- **Genetic testing (definitive):**
- **Trio whole-exome (WES) or whole-genome (WGS) sequencing** to detect the *de novo* CLCN7 missense variant (approach used in all reported cases). Single-gene CLCN7 sequencing or lysosomal/leukodystrophy gene panels including CLCN7 are alternatives.
- **Functional confirmation:** electrophysiology (*Xenopus* oocyte currents) demonstrating GoF; lysosomal pH measurement showing hyperacidity (research setting).
- CMA/karyotype/FISH/mtDNA/repeat testing: not indicated (point-mutation disease).
- **Clinical/laboratory:**
- Ophthalmologic exam (iris transillumination, nystagmus, foveal hypoplasia, VEP) for albinism.
- Abdominal ultrasound/MRI for hepatosplenomegaly.
- **Brain MRI:** delayed/deficient myelination, white-matter signal changes.
- CBC/blood smear: vacuolated lymphocytes; immunoglobulin panel (hypogammaglobulinemia).
- Skin/marrow biopsy or EM: lysosomal storage material/vacuolization (research/confirmatory).
- **Differential diagnosis:** Oculocutaneous albinism (OCA1–7; distinguished by absence of organomegaly/CNS storage), Chediak-Higashi & Hermansky-Pudlak syndromes (lysosome-related organelle disorders with pigment dilution — distinguished by giant granules/bleeding/immune profile and their own genes), other lysosomal storage/leukodystrophies, **CLCN7 osteopetrosis** (distinguished by increased bone density — absent in HOD).

LOINC/ontology suggestions: genetic sequencing panels; brain MRI (RadLex).

11. Outcome / Prognosis

- **Survival/mortality:** No formal survival statistics (too few cases). Reported patients have survived into childhood; long-term prognosis is guarded owing to neurodevelopmental disability and potential progressive neurodegeneration (extrapolated from CLC-7-deficiency models).
 - **Morbidity/function:** Significant lifelong disability — visual impairment, developmental/intellectual disability, and infection susceptibility (hypogammaglobulinemia).
 - **Complications:** Recurrent infections (immune), visual complications of albinism, feeding/growth issues.
 - **Prognostic factors:** Severity of developmental delay and extent of CNS myelination deficit; degree of immune involvement.
 - **Prognostic biomarkers:** None validated; lysosomal storage burden and MRI myelination are candidate markers.
-

12. Treatment

No disease-specific or curative therapy exists; management is supportive.

- **Supportive/rehabilitative (mainstay):** Developmental/early-intervention therapy, physical/occupational/speech therapy (NCIT: rehabilitation therapy); low-vision aids and ophthalmologic management for albinism; photoprotection (sun protection, given hypopigmentation → skin-cancer risk).
- **Immune:** Immunoglobulin replacement (IVIG) and infection prophylaxis if clinically significant hypogammaglobulinemia (NCIT: Intravenous Immunoglobulin Therapy).
- **Organomegaly/storage:** Monitoring; no approved substrate-reduction/enzyme therapy for this entity.
- **Pharmacogenomics:** None established.
- **Mechanism-rational (experimental/hypothetical):** Because the defect is lysosomal **hyper-acidification** from CLC-7 GoF, agents that **raise lysosomal pH / reduce CLC-7 or V-ATPase activity** are conceptually attractive but **untested** in patients. CLC-7 inhibitors are in preclinical study for other indications. No gene-, cell-, or RNA-based therapy has been reported for HOD.

- **Advanced therapeutics (osteopetrosis contrast):** Hematopoietic stem-cell/bone-marrow transplant is standard for *recessive* osteopetrosis and studied for ADO2 in mice

(
P 35959867)

— **not applicable to HOD**, which lacks the osteoclast/bone phenotype.

- **Clinical trials:** None specific to HOD identified on ClinicalTrials.gov (no NCT).

13. Prevention

- **Primary prevention:** Not possible (de novo mutation); no modifiable risk factors.
- **Secondary prevention:** Early diagnosis via trio sequencing enables early developmental intervention, vision support, and immune monitoring.
- **Genetic counseling:** Recurrence risk for parents is low (de novo), but **germline mosaicism** cannot be excluded → prenatal/preimplantation testing can be offered for known familial variants. Affected individuals would have 50% transmission risk (autosomal dominant), though reproduction is unlikely given severity.
- **Screening:** Not part of newborn/carrier screening panels. Cascade testing not generally indicated for de novo cases.
- **Public-health/immunization/prophylaxis:** Standard vaccination and infection prophylaxis if immunodeficient.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** CLCN7 is highly conserved. Orthologs: mouse *Clcn7* (NCBI Gene 26373; NCBI Taxon 10090), rat *Clcn7*, zebrafish *clcn7* (Taxon 7955). Partner *Ostm1* similarly conserved.
- **Natural disease in animals:** Naturally occurring **CLCN7/OSTM1 loss-of-function osteopetrosis** exists (e.g., the mouse *grey-lethal* (*gl*, *Ostm1*) and *Clcn7* mutants; cattle/dog osteopetrosis — OMIA entries for CLCN7-related osteopetrosis). No naturally occurring **gain-of-function HOD** counterpart is documented in animals.
- **Comparative biology:** Loss of CLC-7/*Ostm1* recapitulates lysosomal storage + neurodegeneration + osteopetrosis across species, confirming conserved lysosomal/

osteoclast roles; the *grey-lethal* mouse also shows **coat-color/pigment dilution**, echoing the pigment link.

- **Zoonotic potential:** None (genetic disease).

15. Model Organisms

- **Mouse (primary):**

- **Clcn7^{-/-} knockout** — severe lysosomal storage disease + neurodegeneration (NCL-like, subunit-c and lysosomal-enzyme accumulation), retinal degeneration, osteopetrosis; osteoclast-rescue transgene extends life but CNS disease progresses

(
P 15706348
).

Recapitulates the storage/neurodegeneration axis of HOD (but represents LoF, so bone phenotype opposite).

- **Ostm1 (grey-lethal) mutant** — phenocopy of Clcn7 loss (lysosomal storage, neurodegeneration, osteopetrosis, coat-color dilution).

- **ADO2 knock-in (p.G213R)** — dominant-negative bone model

(
P 35959867
).

— informs CLCN7 dominant biology though not the GoF HOD mechanism.

- **Knock-in of HOD GoF alleles (e.g., Y99C/Y715C)** — appropriate model to recapitulate hyperacidification/hypopigmentation; described in the 2024 GoF characterization work (Polovitskaya et al.,

P 38838776
).

- **Zebrafish:** *clcn7* and *ostm1* mutants show microglial/macrophage phagocytic failure (impaired clearance of apoptotic cells and β -amyloid) with normal microglial development

(
P 38294065
).

— models the neuroimmune/clearance component.

- **In vitro/heterologous:** *Xenopus* oocyte and mammalian-cell expression for electrophysiology and lysosomal-pH assays (functional variant validation;

P 31155284

P 38136669

P 33125761

Arabidopsis vacuole system for CLC-7 transport mechanism

P 22641774

- **Model limitations:** Mouse KO models loss-of-function, not the human GoF; GoF knock-ins and patient iPSC-derived neurons/melanocytes are needed to fully model HOD-specific hyperacidification and hypopigmentation.
- **Resources:** MGI (*Clcn7*, *Ostm1*), ZFIN (*clcn7*), IMPC/IMSR for alleles; OMIA for animal osteopetrosis.

Key References (PMID)

- **31155284** — Nicoli et al. 2019, *Am J Hum Genet*. Original description: de novo CLCN7 variant → lysosomal storage + albinism via altered lysosomal acidification (the defining HOD paper).
- **38838776** — Polovitskaya et al. 2024. Gain-of-function CLCN7 variants cause hypopigmentation and lysosomal storage disease (mechanistic/model confirmation).
- **39056574** — Lee et al. 2024. Multisystem CLCN7 p.Y715C disorder (developmental delay, organomegaly, hypogammaglobulinemia, hypopigmentation) **without osteopetrosis**; GoF/lysosomal hyperacidity.
- **38136669** — Bose et al. 2023. Impaired autophagic clearance with a GoF CLC-7 variant.
- **32749217** — Schrecker et al. 2020. Cryo-EM of CLC-7/OSTM1 complex.
- **15706348** — Kasper et al. 2005. *Clcn7*-KO mouse: lysosomal storage disease + neurodegeneration.
- **38294065** — Iyer & Talbot 2024. CLC-7 essential for microglial phagocytic clearance (zebrafish).
- **33125761** — Di Zanni et al. 2021. Structure/function of 14 CLC-7 mutants; lysosomal localization ↔ neurodegeneration.
- **11741829** — Cleiren et al. 2001. ADO2 = dominant-negative CLCN7 (contrast to GoF HOD).
- **17936098** — Del Fattore et al. 2008. Osteopetrosis genetics/pathogenesis review.

- **35959867** — Alam et al. 2022. BMT for ADO2 in mice.
- **11207362** — Kornak et al. 2001. CLC-7-deficient mice: osteopetrosis + retinal degeneration; CLC-7 provides Cl^- conductance for osteoclast H^+ -ATPase.
- **16525474** — Lange et al. 2006. CLC-7 requires OSTM1 β -subunit; grey-lethal (Ostm1-null) mice phenocopy Clcn7-null (storage, neurodegeneration, osteopetrosis).
- **33217309** — Polovitskaya et al. 2020. Recurrent gain-of-function CLCN6 (CLC-6) mutation → neurodevelopmental/neurodegenerative disorder (endolysosomal CLC GoF paradigm).
- **33708769** — Bose, He & Stauber 2021. Review: neurodegeneration upon dysfunction of endosomal/lysosomal CLC transporters.
- **12059962** — Yoshikawa et al. 2002. CLC-3 deficiency → NCL-like phenotype with subunit-c accumulation and elevated endosomal pH.
- **22335796** — Mindell 2012. Lysosomal acidification mechanisms (V-ATPase + CLC-7 counterion).
- **31214276** — Miao et al. 2019. Intramelanocytic acidification suppresses tyrosinase/melanogenesis.
- **19389708** — Cheli et al. 2009. Melanosome pH as key regulator of melanin synthesis (α MSH/cAMP).

Consolidated Causal-Chain Synthesis (knowledge-base ready)

One defect, four tissue readouts. A *de novo* gain-of-function *CLCN7* missense variant → increased CLC-7 $2\text{Cl}^-/1\text{H}^+$ antiporter activity → excess counterion shunt for the V-ATPase → **over-acidification** of lysosomes and lysosome-related organelles → branches: - **Melanosome branch** → tyrosinase (pH-sensitive, minimal activity when acidic) suppressed → reduced melanin → **hypopigmentation / albinism** (HP:0001010; evidence: in vitro melanosome-pH biology + human). - **Autophagy/visceral branch** → hydrolase pH-mismatch → impaired autophagic/lysosomal clearance → storage material + vacuolization in liver/spleen macrophages → **organomegaly** (HP:0003271; evidence: human cells + in vitro). - **Neural branch** → neuronal/oligodendrocyte/microglial lysosomal dysfunction and NCL-like storage → **delayed myelination + developmental delay** (HP:0012448, HP:0001263; evidence: model organism + human). - **Immune branch** → impaired endolysosomal processing in B-lineage cells → **hypogammaglobulinemia** (HP:0004313; evidence: human case). - **Spared:** osteoclast bone resorption (requires *reduced* CLC-7) → **no osteopetrosis** (the discriminating negative feature).

Confirmed Findings Recorded (6)

1. HOD (OMIM 618541) = de novo **gain-of-function** *CLCN7* — mirror image of loss-of-function/ dominant-negative osteopetrosis [human + in vitro;

P 39056574

P 31155284

P 11741829

2. CLC-7/OSTM1 dysfunction → lysosomal storage + NCL-like neurodegeneration; obligate OSTM1 complex [model organism + structural;

P 15706348

P 32749217

P 16525474

P 33125761

3. Clinical triad hypopigmentation + organomegaly + delayed myelination/development **without osteopetrosis** [human;

P 39056574

4. Hypopigmentation via melanosome over-acidification suppressing tyrosinase [in vitro/ human;

P 31214276

P 19389708

P 22335796

5. Organomegaly via impaired autophagic clearance and reticuloendothelial storage [in vitro/ human;

P 38136669

P 39056574

6. Delayed myelination/development via neural + microglial lysosomal failure;
endolysosomal-CLC GoF paradigm (CLCN6) [model organism + human;

P 16525474

P 11207362

P 38294065

P 33217309

P 33708769

Evidence-Source Classification

- **Human clinical:**

P 31155284

(Nicoli 2019, original probands), 39056574 (Lee 2024), 11741829 (osteopetrosis genotype-phenotype).

- **Model organism:**

P 15706348

& 11207362 (mouse Clcn7-KO), 16525474 (Ostm1 grey-lethal mouse), 38294065 (zebrafish clcn7/ostm1), 35959867 (ADO2 mouse).

- **In vitro / functional / structural:**

P 38136669

(autophagy), 33125761 (14 mutants), 32749217 (cryo-EM), 22335796/31214276/19389708 (pH/melanosome biology), 36641477 (TPC2 albinism), 18836173 (v-ATPase zebrafish).

- **Review/computational:**

P 33708769

P 17936098

P 35159175

Limitations

- Evidence base is very small (individual case reports + functional/model studies); frequencies, prognosis, survival, and QoL metrics are qualitative.
- Some mechanistic links (melanosome hyperacidification → hypopigmentation; storage → organomegaly; progressive neurodegeneration) are **inferred** from lysosome/melanosome biology and LoF models rather than directly demonstrated in HOD patients.
- No approved therapy; mechanism-based treatments remain hypothetical.
- Ontology cross-references (Mondo, ICD-11, Orphanet) should be verified against current releases.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery